

The history of beer additives in Europe – a review

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Abstract. The many excavations of medieval sites during recent years have resulted in a strong increase in archaeobotanical records including species which were used as beer additives. Since the first compilation of records by the author in 1984 relating to the two main species, namely *Myrica gale* and *Humulus lupulus*, the number of finds has quadrupled. Distribution maps of the sites with fossil occurrence of these two species are presented and this evidence is complemented by that from written sources. *M. gale* seems to have been used for brewing as early as the centuries immediately before and after the birth of Christ in a small area at the Rhine estuary in the northern Netherlands. During the early and high Middle Ages there are records of this plant, in what are potentially brewing contexts, across its north-west European area of natural distribution. Written sources confirm its use in brewing as early as the tenth century. The finds of *H. lupulus* indicate that this species has been used in brewing from the early Middle Ages and this hypothesis is supported by documentary evidence. Cultivation of hop began around A.D. 859. In the late Medieval period, strong competition developed between both kinds of beer, which resulted in the take-over by *H. lupulus* in the eighteenth century. Many other herbs of secondary importance have been used to flavour beer or to prepare medicinal beers. These are mentioned in old herbals and have been compiled in this paper. These various flavouring agents, combined with the use of all available species of cereals led to a variety of beers that is unimaginable today.

Key words: Beer additives – Brewing – *Myrica gale* – *Humulus lupulus* – Middle Ages

Introduction

In contrast to wine which is made from sugars derived from fruit that often have distinctive tastes, starch is the starting point for beer production, and hence cereals are mainly used in brewing. Starch, however, does not have much taste, so additional flavouring agents are necessary

to improve the quality and taste of beer. Apart from flavour, the alcohol content of beer is much lower than that of wine, and hence beer is more perishable. Preservatives are necessary to increase the stability or 'shelf-life' of beer and selected plants have been used in the past for this purpose.

Firstly, a brief review of the history of beer itself is appropriate. Beer production is depicted in reliefs, paintings and inscriptions from Mesopotamia and Egypt from as early as Neolithic times. There is evidence that malt from barley and emmer and additives such as dates and other herbs, the identity of which is unknown, were used (Helck 1971). Beer was known in ancient Greece and Rome, but wine has been the preferred drink in these regions. Beer, however, was the main drink of the Germanic and Celtic tribes as has been recorded in Roman sources, in particular by Tacitus. But there is also considerable archaeobotanical evidence for sprouted cereal grain mainly from Roman sites in north-west Europe (cf. van Zeist 1991) and there are records also from a Celtic context (Hochdorf) in southern Germany (Stika 1996).

In former times most cereals were used for malting. Apart from barley, which is to-day the cereal preferred by the brewing industry, emmer, oats, spelt wheat, bread wheat were widely used and, in Estonia, rye was used up until the nineteenth century (Moora 1991). Outside Europe, millet, rice, maize and tuber plants have been, and are still, commonly used.

Another necessary ingredient is yeast as there is no fermentation without it. The role that this fungus played was, however, unknown in older times. Success was therefore dependent on the spores of yeast, which are carried in the air, infecting the brew by chance; otherwise the brew was a failure. This important aspect was recognised only much later. It is first mentioned as late as 1551 in Munich.

The topic of the paper presented here is the use of additives to flavour beer and as preservative agents. There are only a few written sources from the early and high Middle Ages that indicate that certain plants were used to improve the beer. These occasional passages, however, are very important in that they serve to shed light on the evidence from archaeobotanical sources.

Otherwise, it is difficult to establish beyond doubt that remains recovered from archaeological contexts are indeed connected with brewing. In this review, only those species that are mentioned in written sources, in the context of beer and beer-making, are regarded as beer additives or beer spices. Species, other than those referred to here, may also have been used during particular periods or in regions where there are few written records.

Some 15 years ago, a first compilation of the occurrence of the most important beer additives was published by the author (Behre 1984). Due to the considerable progress in archaeological and archaeobotanical studies of the Medieval period, the available fossil evidence has quadrupled since that time and so gives a solid basis for charting the history of the various relevant species.

Plants used in the past as beer additives

Among the many species which have been used as beer additives, two plants were of special importance, namely, sweet gale or bog myrtle (*Myrica gale* L.) and hops (*Humulus lupulus* L.). One or other of these was used in the case of all beers in Europe during the Middle Ages and later. They were complemented by other species to give a variety of special flavours to beer. As will be shown below, there are numerous records of *M. gale* and *H. lupulus* from Medieval and post-Medieval excavations. Some of the finds are so numerous or the detailed context is such that the conclusion that they were used for brewing is inescapable. However, not every single record of the fruitlets of sweet gale or hops points to a brewery as both species belong to the natural flora of rather extensive parts of Europe. Records from contexts such as raised bogs, e.g. *M. gale* recovered with other raised bog taxa such as *Sphagnum* and *Calluna*, or from burials, where hop is known to have been used as bedding material for the corpse, are not included.

It is important to note that the remains of both *M. gale* and *H. lupulus* are generally found uncarbonized. Charred finds of fruitlets of sweet gale or hops are very rare. This means that the fossil evidence is connected with waterlogged sites where fruitlets, leaves, etc., are preserved. Fortunately, many of the Medieval sites that have been investigated yield waterlogged material. It should, however, be borne in mind that the records are probably somewhat skewed insofar as sites on sandy soils outside urban contexts normally yield only carbonized remains.

The archaeobotanical record of sweet gale, Myrica gale L.

Nowadays, it is almost forgotten that in former times *M. gale* played a very important role in the beer industry of several countries. This shrub, referred to as sweet gale or bog myrtle in English and *Gagel* in German and Dutch, grows on acid bogs and sandy soils in north-west Europe. It is an oceanic species and its distribution area includes the northernmost parts of the central European mainland as well as north-west Europe and the coastal

zone of the Baltic Sea (Fig. 1). All parts of this plant have an aromatic smell due to the gale oil which is excreted through glands in the leaves and flowers.

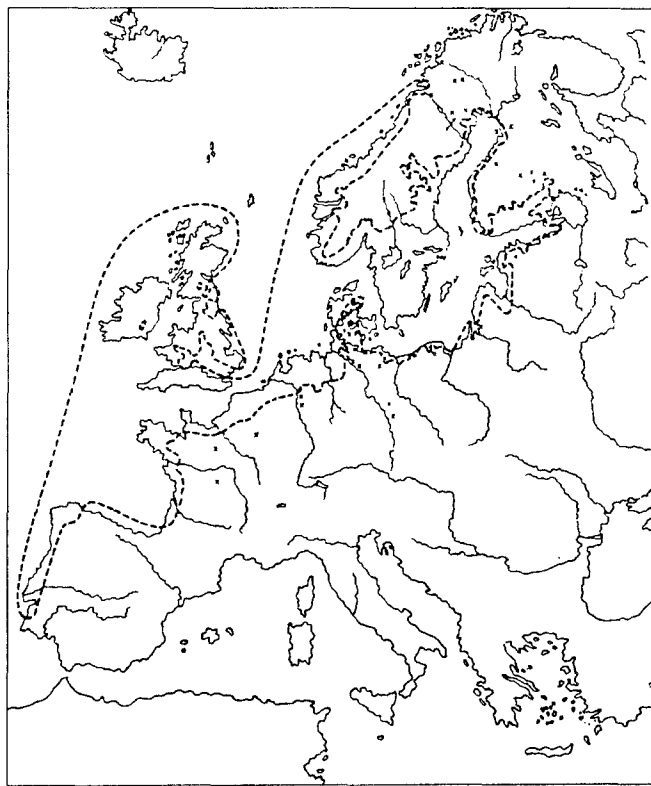


Fig. 1. Natural distribution area of sweet gale, *Myrica gale* (after Meusel, from Hegi 1957)

Twigs, leaves and fruitlets as well as pollen are commonly recorded in the context of palaeoecological investigations of peat deposits within the present-day range of *M. gale*. These numerous records from natural contexts are excluded from the distribution map (Fig. 3) which is based exclusively on records from archaeological sites and consists mainly of fruitlets and sometimes leaves.

In some excavations, the remains of *M. gale* are so abundant that it is clear that they were intended for beer making. This is the case, for example, at 'Alte Boom-

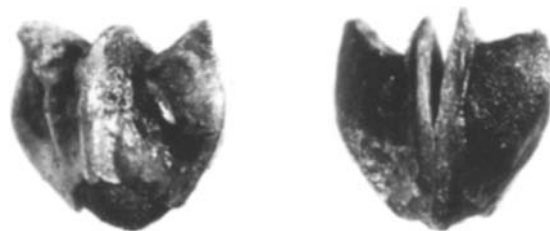


Fig. 2. Subfossil *Myrica gale* from Haithabu, ninth century (left) and Bremen, thirteenth century (right); both 15:1

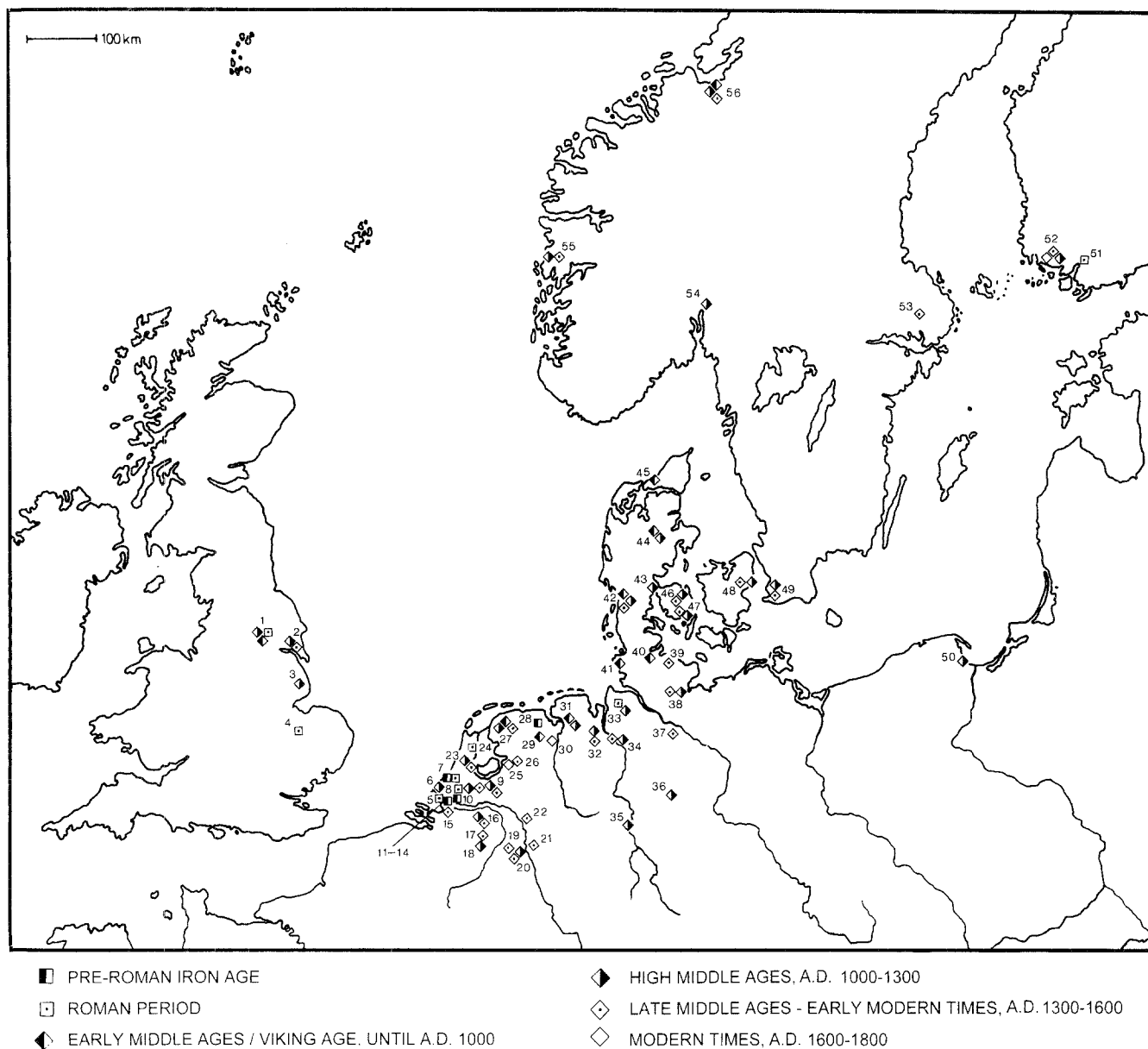


Fig. 3. Archaeological sites with macro-remains of sweet gale, *Myrica gale* L.

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|---|--|--|
| 1 York, Hall and Kenward 1990;
Kenward and Hall 1995 | 22 Wesel, Knörzer et al. 1995 | 39 Kiel, Wiethold and Schulz 1991 |
| 2 Beverley, Hall 1992; McKenna 1992 | 23 Amsterdam, Pals 1972b;
Paap 1983 | 40 Haithabu, Behre 1983 |
| 3 Lincoln, Greig 1993 | 24 Schagen, van Haaster 1985 | 41 Elisenhof, Behre 1976 |
| 4 Stonea, van der Veen 1996 | 25 Kampen, Brinkkemper and
Vermeeren 1997 | 42 Ribe, Bencard and Lange 1972; Jensen 1986;
Robinson 1994 |
| 5 Nieuwenhoorn, Brinkkemper 1993b | 26 Westenholte, van Vilsteren 1983 | 43 Kolding, Jensen 1986 |
| 6 Leiden, Kuijper 1982 | 27 Leeuwarden, van Zeist et al. 1987 | 44 Viborg, Jensen 1986; Robinson and
Boldson 1998 |
| 7 Valkenburg, Brinkkemper unpubl. | 28 Groningen, van Zeist 1974 | 45 Aggersborg, Jessen 1954 |
| 8 Rotterdam, Brinkkemper 1997a, b | 29 Peelo, van Zeist and
Palfenier-Vegter 1995 | 46 Odense, Jensen 1986 |
| 9 Amersfoort, Hänninen and de Man 1994 | 30 Bourtange, van Zeist 1993 | 47 Svendborg, Jensen 1979; Jørgensen 1986 |
| 10 Vlaardingen, van Zeist 1974 | 31 Hatzum, Behre 1986 | 48 Copenhagen, Rostrup 1906 |
| 11 Zuidland, Brinkkemper 1993b | 32 Oldenburg, Kučan 1998 | 49 Lund, Hjelmqvist 1991 |
| 12 Abbenbroek, Brinkkemper 1993b | 33 Flögel, Behre and Kučan 1994 | 50 Gdansk, Klichowska 1972 |
| 13 Geervliet, Brinkkemper 1993b | 34 Bremen, Behre 1991 | 51 Salo, Aalto 1982 |
| 14 Spijkenisse, Brinkkemper 1993b | 35 Hörter, Willerdig 1986 | 52 Turku, Aalto 1994; Lempiäinen 1994, 1995 |
| 15 Dordrecht, Pals 1972a | 36 Braunschweig, Hellwig 1990 | 53 Uppsala, Pahlsson unpubl. |
| 16 's-Hertogenbosch, van Haaster 1997 | 37 Lüneburg, Behre 1981 | 54 Oslo, Griffin 1977, 1979 |
| 17 Eindhoven, Luijten 1992, 1994 | 38 Lübeck, Lynch and Paap 1982;
Alsleben 1991 | 55 Bergen, Griffin 1981; Krzywinsky and
Soltved 1988 |
| 18 Dommelen, Pals 1988 | | 56 Trondheim, Tallantire 1979; Griffin and
Sandvik 1989 |
| 19 Brüggen, Knörzer 1979 | | |
| 20 Neuss, Knörzer 1984, 1988 | | |
| 21 Duisburg, Knörzer 1992 | | |

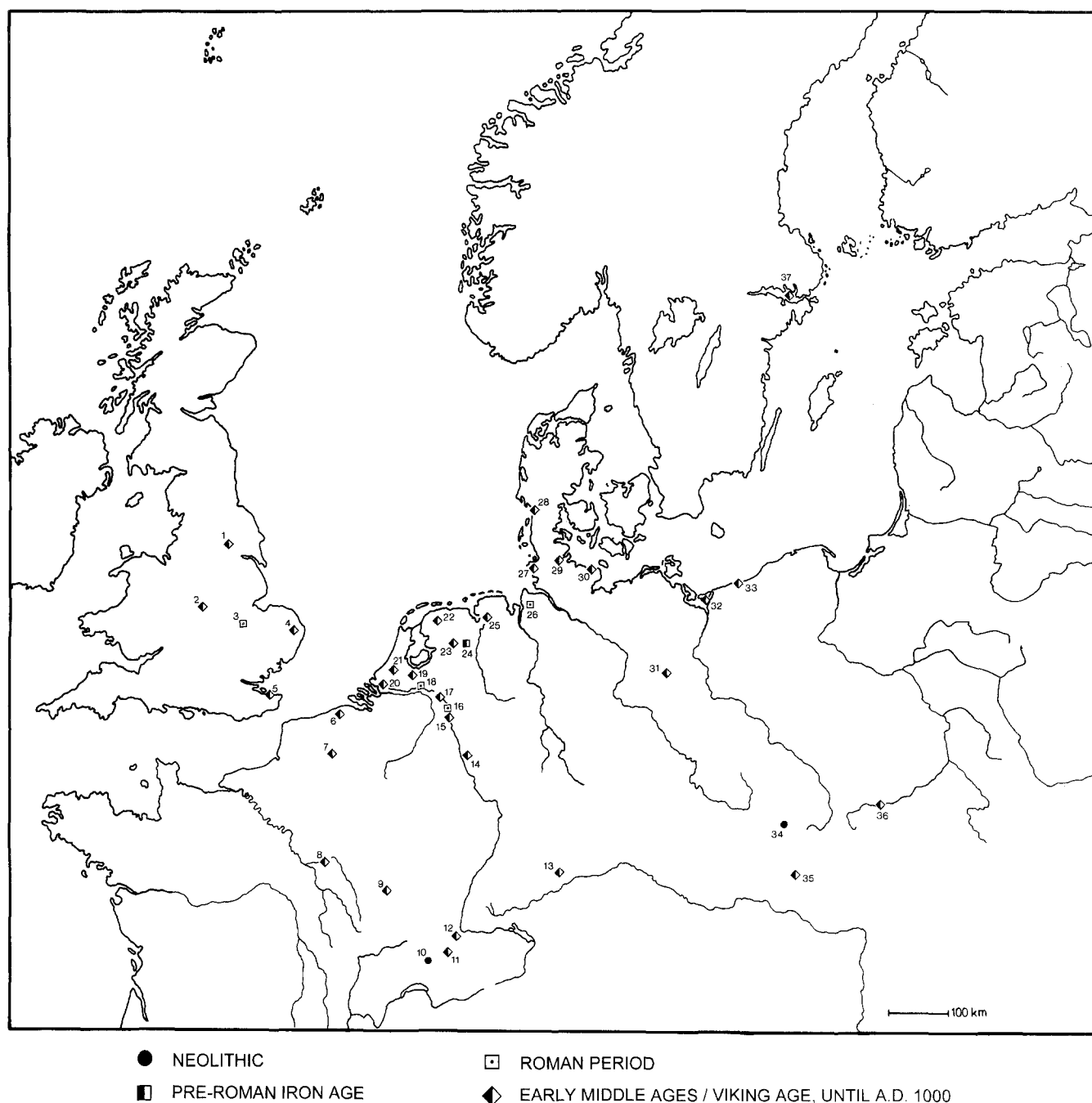


Fig. 4. Archaeological sites relating to the period from the Neolithic until A.D. 1000 with macro-remains of hop, *Humulus lupulus* L. Site names in *italics* are also mapped in Fig. 6

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|--|--------------------------------------|--|
| 1 York, Godwin and Bachem 1961;
Kenward and Hall 1995 | 14 Siegburg, Knörzer 1994 | 29 Haithabu, Behre 1983 |
| 2 Tamworth, Thomas and Greig 1992 | 15 Krefeld-Oppum, Knörzer 1981 | 30 Oldenburg i.H., Kroll 1991 |
| 3 Longthorpe, Wilson 1978 | 16 Xanten, Knörzer 1981 | 31 Spandau, Schumann 1990 |
| 4 Norwich, Murphy 1988, 1994 | 17 Kalkar, Knörzer 1996 | 32 Wolin, Klichowska 1972; Alsleben 1997 |
| 5 Graveney, Wilson 1975 | 18 Wijk bij Duurstede, Lange 1990 | 33 Kołobrzeg-Budzistowo, Klichowska 1972 |
| 6 Zerkegem, Hollevoet et al. 1993 | 19 Houten, Kooistra and Hessing 1988 | 34 Mohelnice, Opravil 1979 |
| 7 Douai, van Zeist et al. 1994 | 20 Rotterdam, Brinkkemper 1997a | 35 Mikulčice, Opravil 1983 |
| 8 Serris-Les Ruelles, de Hingh and
Bakels 1996 | 21 Leiderdorp, Pals 1986 | 36 Kraków, Wasylukowa 1979; Wasylukowa
and Zemanek 1995 |
| 9 Bailleul-en-France, Ruas 1988 | 22 Leeuwarden, van Zeist et al. 1987 | 37 Birka, Hansson 1997 |
| 10 Brise-Lames, Baudais-Lundstrom 1978 | 23 Pesse, van Zeist et al. 1986 | |
| 11 La Neuveville, Brombacher et al. 1997 | 24 Noordbarge, van Zeist 1983 | |
| 12 Develier, Brombacher et al. 1997 | 25 Hatzum, Behre 1986 | |
| 13 Lauchheim, Rösch 1992 | 26 Flögel, Behre and Kučan 1994 | |
| | 27 Elisenhof, Behre 1976 | |
| | 28 Ribe, Jensen 1986; Robinson 1994 | |

borg', the seat of a Frisian nobleman in north-west Germany from the ninth/tenth to the thirteenth century A.D., where large amounts of *M. gale* leaves as well as 469 fruitlets were collected in 12 samples and in a single sample the fruitlets of *H. lupulus* were also recorded (Behre 1986). Here there can be no doubt that sweet gale was employed for brewing. Other large *Myrica* finds include Viborg and Ribe, Denmark (10-13th century, 374 fruitlets; 8-12/13th century, 704 fruitlets; both records in Jensen 1986), as well as Lund, Sweden (11-15th century, 3256 fruitlets, Hjelmqvist 1991) and Lincoln, England (10-12th century, 75 fruitlets, Greig 1993).

Noteworthy is the small number of rather early records for *M. gale* (Fig. 3). These include seven sites from the pre-Roman Iron Age, all of which relate to the Netherlands. Apart from site 28, they all yielded several remains, e.g. up to 108 bud fragments at Geervliet. The sites are concentrated in the northern part of the Rhine estuary and all come from clear archaeological contexts such as pits, settlement layers, etc. Rockanje, another pre-Roman Iron Age site in the area from this period and with large quantities of *M. gale* remains (Brinkkemper 1993) has been excluded, because these finds clearly derive from the natural peat beneath the settlement.

The regular occurrence, the clear archaeological context and the number of finds lead to the conclusion that sweet gale was already in use during the pre-Roman Iron Age, presumably for making beer. However, this relates to a rather limited area. From the subsequent Roman period, there are eight sites with sweet gale, four of them from the same area. Sites 5 and 8 yielded not only fruitlets (up to 609 in Nieuwenhoorn), but also leaves and buds and so confirm the assumption that sweet gale was used here on a continuous basis. Apart from sites 33 and 51, where only single fruitlets were found (which is not very informative), sites 3 and 51 (from Britain and Finland, respectively) yielded 11 and 31 fruitlets, respectively. At both sites, charred specimens were recorded and even if there is no direct evidence for brewing activities, the possibility cannot be excluded.

During the Middle Ages the numbers of records decline. Ten finds, some of them with large numbers of specimens, date back to before A.D. 1000 and many more are from the high Middle Ages until modern times. This is in agreement with the oldest written sources which will be discussed later and so there is no doubt that most of the *Myrica* finds from archaeological sites are indeed connected with brewing. The site distribution map shows that in Medieval times, *M. gale* was used as a beer additive over an area that coincides with the natural distribution of this species (Fig. 3). From outside this area there are only few finds (cf. site 35, Hörter, which lies, at most, 60 km south of its present distribution). The use of sweet gale was therefore more or less strictly limited to where it grew naturally and long-distance trade in this beer additive does not arise.

The archaeobotanical record for hops, Humulus lupulus L.

The natural distribution area of *H. lupulus* includes the temperate regions of Europa and extends to mid-Scandinavia as well as the Mediterranean region. Hop is a

climber and occurs naturally in flood-plain forests and carr, i.e. fen woods. After the opening-up of the natural forests in the mid-Holocene, it found favourable habitats along forest edges and also colonised man-made habitats such as hedges and buildings.

Hop is dioecious. The female cone-like infructescences are used in brewing. The bracts have glands which contain the hop bitter agents lupulone and humulone that flavour the beer and are important also as preservatives. Nowadays, only cultivated hop is used for brewing and, over the last hundred years, only female plants are used that produce neither fruitlets nor pollen.

In waterlogged sites, hop fruitlets are generally well preserved while they are very rare in carbonized material. It is not possible to distinguish fruitlets from wild and cultivated plants. The specific context of each find must therefore be considered. Since *H. lupulus* may occur close to settlements, finds of single or few fruitlets do not necessarily point to their use in brewing. At several sites, however, including sites of an early date, large quantities of hops have been recovered and these can only be explained by invoking brewing. These include sites such as Haithabu, N. Germany (3321 fruitlets, Viking age, Behre 1983), the Graveney boat, England (411 fruitlets, 10th century, Wilson 1975), Wolin, Poland (1640 fruitlets, early to late Medieval period, Klichowska 1972), Lübeck, Germany (1558 fruitlets, 13th century, Lynch and Paap 1982), and Opava (4327 finds, 13-18th century, Opravil 1969-1996).

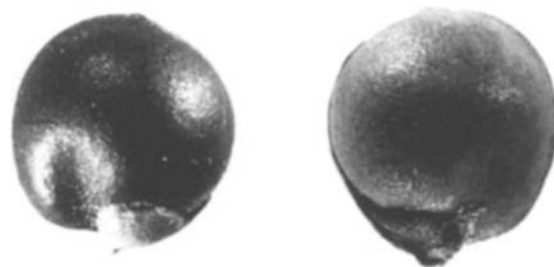


Fig. 5. Subfossil *Humulus lupulus* from Haithabu, ninth century; 15:1

The new compilation includes so many sites for hops that separate maps are used for sites dating to the interval Neolithic to A.D. 1000 (Fig. 4) and from A.D. 1000 onwards (Fig. 6). There are no records available from south of the Alps, which strongly suggests that wine was the main alcoholic drink in those parts.

Figure 4 shows only a few records from prehistoric periods. From the Neolithic, pre-Roman Iron Age and the Roman period, there are two, one and four records, respectively. Apart from one site where two specimens were recorded, the records consist of a single fruitlet. This obviously does not constitute cogent evidence for the use of hops in brewing during prehistoric times.

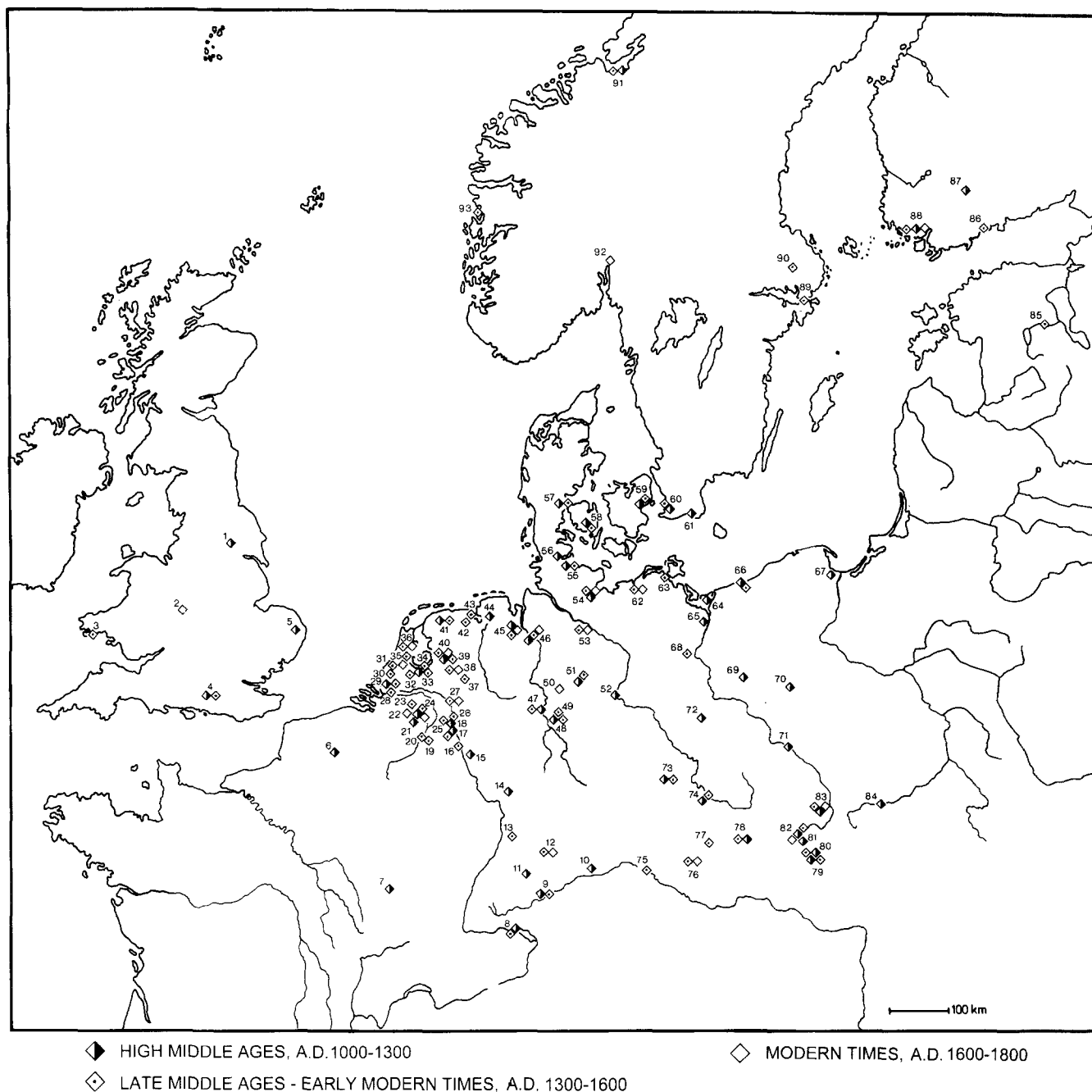


Fig. 6. Archaeological sites dating to period A.D. 1000 to sixteenth century with macro-remains of hop, *Humulus lupulus* L. Site names in *italics* are also mapped in Fig. 4 (for numbers see opposite page)

A sharp increase in numbers of sites takes place in the early Middle Ages. There are thirty finds relating to the Middle Ages before A.D. 1000 (Fig. 4). This compared with only seven single finds from the prehistoric period. Moreover, at several sites, including York (site 1), Graveney (site 5), Develier (site 12), Haithabu (site 29), Wolin (site 32), Kołobrzeg (site 33) and Mikulčice (site 35) more than a hundred, and frequently up to several thousand, fruitlets have been recorded (site numbers are as in Fig. 4). The average number of fruitlets per site jumps from 1.3 in Roman times to 209.9 during the early Middle Ages. This sharp rise in the quantity of hops at individual sites points to intentional and intensive use of

hops, and this can only signify that hops were being used for brewing. The oldest early Medieval finds of *H. lupulus* are dated earlier than A.D. 800 (Develier, site 12, 175 specimens, 6-8th century), Serris-Les Ruelles (site 8, no numbers given, 7-9th century) and Mikulčice (site 35, 235 specimens, 8-9th century). Other important early finds include Haithabu (site 29, 3321 specimens, 9-10th century) and, in particular, the famous Graveney boat (site 5, 411 fruitlets and 136 bracteoles, 10th century). The Graveney-boat find shows that hops were already being traded in this early period and this can only be explained in the context of brewing.

Locations of Fig. 6.

- 1 York, Kenward and Hall 1995
- 2 Dudley, Moffet 1992
- 3 Tenby, Murphy et al. 1989
- 4 Alton, Green 1983
- 5 Norwich, Murphy 1988
- 6 Douai, van Zeist et al. 1994
- 7 Baillet-en-France, Ruas 1988
- 8 Konstanz, Küster 1989, 1992
- 9 Ulm, Wiethold 1993
- 10 Eichstätt, Küster 1992
- 11 Gerlingen, Rösch and Gross 1994
- 12 Schwäbisch Hall, Rösch et al. 1994; Rösch and Fischer 1995
- 13 Heidelberg, Maier 1983
- 14 Oberursel, Kreuz 1993
- 15 Siegburg, Knörzer 1994
- 16 Köln, Knörzer 1991b
- 17 Neuss, Knörzer 1984, 1988; Knörzer and Müller 1968
- 18 Buderich, Knörzer 1968
- 19 Sittard, Bakels 1980
- 20 Maaseik, van den Brink 1989
- 21 Dommelen, Pals 1988
- 22 Tilburg, Pals-van Blerk 1986
- 23 's-Hertogenbosch, van den Brink 1988; van Haaster 1997
- 24 Eindhoven, Luijten 1992, 1994
- 25 Wachtendonk, Knörzer 1991a
- 26 Duisburg, Knörzer 1992
- 27 Wesel, Knörzer et al. 1995
- 28 Dordrecht, Pals 1972a
- 29 Rotterdam, Brinkkemper 1997b
- 30 Zoetermeer, Vijverberg 1993
- 31 Leiden, Kuijper 1985
- 32 Utrecht, van den Brink 1984
- 33 Amersfoort, Hänninen and de Man 1994
- 34 Harderwijk, Pals 1980
- 35 Amsterdam, Pals 1972b; Paap 1983
- 36 Alkmaar, Esser and Gehasse 1995
- 37 Enschede, Buurman 1976
- 38 Deventer, Botanisch. Lab. 1982
- 39 Westenholte, van Vilsteren 1983
- 40 Kampen, Vermeeren 1990; Brinkkemper and Vermeeren 1997
- 41 Leeuwarden, van Zeist et al. 1987
- 42 Groningen, van Zeist 1992
- 43 Heveskesklooster, Cappers 1995
- 44 Hatzum, Behre 1986
- 45 Oldenburg i.O., Kučan 1998
- 46 Bremen, Behre 1991
- 47 Höxter, Willerding 1983, 1986; Wolf 1991
- 48 Göttingen, Willerding 1987; Hellwig 1997
- 49 Northeim, Hellwig and Kuprat 1991
- 50 Hildesheim, Willerding 1990
- 51 Braunschweig, Hellwig 1990
- 52 Magdeburg, Lange 1987
- 53 Lüneburg, Behre 1981; Wiethold 1995a, 1996
- 54 Lübeck, Paap 1984; Lynch and Paap 1986; Averdick 1990; Alsleben 1991
- 55 Kiel, Heinrich et al. 1994; Wiethold 1995b
- 56 Schleswig, Pasternak 1991
- 57 Kolding, Jensen 1986
- 58 Svendborg, Jensen 1979; Jørgensen 1986
- 59 Copenhagen, Rostrup 1906
- 60 Lund, Hjelmqvist 1991, 1995
- 61 Ilstorp, Hjelmqvist 1992
- 62 Rostock, Wiethold unpubl.
- 63 Stralsund, Wiethold unpubl.
- 64 Wolin, Klichowska 1972; Alsleben 1997
- 65 Szczecin, Klichowska 1972
- 66 Kołobrzeg, Klichowska 1972; Latałowa and Badura 1996
- 67 Gdansk, Klichowska 1972
- 68 Eberswalde, Wiethold unpubl.
- 69 Miedzyrzecz Wlkp., Klichowska 1972
- 70 Bnin, Klichowska 1975
- 71 Wrocław, Kosina 1977, 1978, 1995
- 72 Cottbus, Lange 1993
- 73 Most, Opravil 1979; Čulíkova 1995
- 74 Prague, Opravil 1994; Čulíkova 1987
- 75 Deggendorf, Küster 1988
- 76 Prachatitz, Suchá and Kočar 1996
- 77 Tabor, Opravil 1985a
- 78 Jihlava, Kühn 1991
- 79 Uherský Brod, Opravil 1993
- 80 Uherské Hradiště, Opravil 1985b
- 81 Prešov, Opravil 1990a
- 82 Olomouc, Opravil 1984, 1985, 1990b, 1994
- 83 Opava, Opravil 1969, 1986, 1990c, 1993, 1995, 1996
- 84 Kraków, Gluza 1970; Wasylikowa 1979
- 85 Tartu, Sillasoo 1996
- 86 Helsinki, Vuorela and Lempiäinen 1993
- 87 Varikkoniemi, Lempiäinen 1992
- 88 Turku, Lempiäinen 1985, 1988, 1994, 1995; Aalto 1994
- 89 Stockholm, Griffin 1982
- 90 Uppsala, Pahlsson unpubl.
- 91 Trondheim, Griffin and Sandvik 1989; Sandvik 1995
- 92 Oslo, Griffin 1981
- 93 Bergen, Krzywinski and Soltved 1988

It is noteworthy that early Medieval hops finds are scattered over large parts of central and western Europe (Fig. 4). To date, there are too few early finds to pinpoint a particular area in which hops might have first been used in brewing. The earliest available records, however, relate to western Switzerland and France.

After A.D. 1000, the number of sites with *H. lupulus* records increases once again as is shown in Figs. 6 and 7. Finds are now common in Medieval sites in central Europe as well as in the west and north, and also the number of fruitlets per site is often high (Fig. 7). There are many records from the Netherlands, N. Germany, and the Czech Republic, while there are large gaps in other regions (Fig. 6). Although the distribution map gives a good impression of the spread in the use of hops for brewing, one has to consider possible deficiencies that will be discussed in the context of all maps (see Conclusion). One point with regard to the sites them-

selves should be emphasized here: beer production in earlier times was not dominated by monasteries as it is often thought today. Most records of hops in the early and later Middle Ages do not come from monasteries. Some of the early northern finds date back to before Christianization in the respective region.

In summary then, there are only a few single finds of *H. lupulus* from prehistoric contexts. In the early Middle Ages there is a sharp rise in numbers of sites and quantities of remains. This can be explained only by assuming that hops were now being used in brewing.

Written sources for the use of sweet gale (M. gale) and hops (H. lupulus) in brewing

Nowadays, sweet gale is no longer used for making beer. There is, however, a long tradition in the Netherlands, N. Germany, S. Scandinavia and England that sweet gale

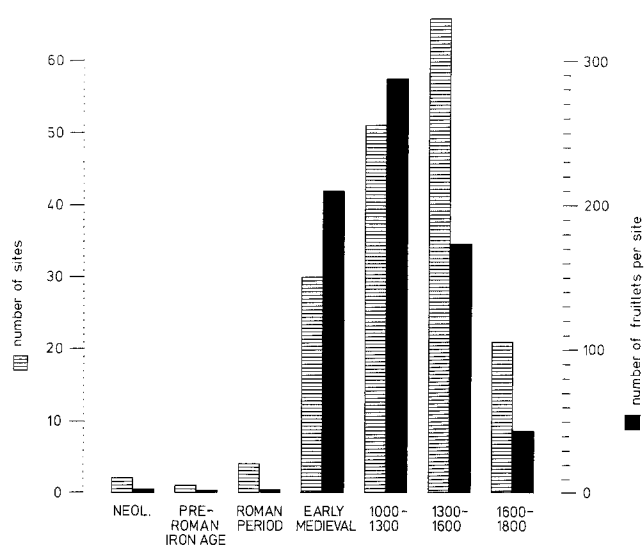


Fig. 7. Number of sites and average numbers of fruitlets per site for *Humulus lupulus* during prehistoric, medieval and modern times

acted as the main ingredient in beer, i.e. apart from malt, and this is documented in many written sources. The oldest go back as far as early Medieval times and have recently been compiled by van Vilsteren (1994). The common name for the beer agent, made mainly from *M. gale*, in the Netherlands and N. Germany was *gruit* or *grut*, and the trade with *grut* was a royal privilege even in the early Middle Ages. The first certain reference to *grut* in its Latin form, *materiam cervisiae*, is in a charter dating to A.D. 974 and issued by Emperor Otto II, in which the *grut* rights of Fosses (Belgium) were transferred to the church at Liège. The word *grut* is first mentioned in A.D. 999, when Emperor Otto III donated the domaine of Bommel, including trade in *grut* to the church of Utrecht. Hildegard von Bingen (1098-1179) recommended the use of sweet gale for making beer, and she disliked hops for this purpose. Also Albertus Magnus (1193-1280), with reference to Holstein, N. Germany, talked about sweet gale and mentioned its use for beer instead of hops.

From the twelfth century onwards, there are numerous written references to sweet gale known from Denmark (Jensen 1979), N. Germany (Behre 1998) and also the Netherlands (van Vilsteren 1994). Within the area of its natural occurrence, sweet gale was generally used before hops came into use as a beer additive. Until the thirteenth century, *M. gale* was the main beer flavouring agent in this region and it often had a monopoly that was, on occasion, defended by political means. The beer made with sweet gale had a sweet taste. Its main disadvantage was that the beer did not preserve well and hence was unsuitable for transporting. It was therefore always brewed in the region of its consumption.

Since only hops are nowadays used in brewing, its history has been intensively researched. All written sources from the early periods have been evaluated with two main questions in mind, namely, what is the earliest evidence for the use of hops in brewing and when did the cultivation of *H. lupulus* start. A critical evaluation of

the oldest sources relating to hops and its possible use in brewing has been made by Wilson (1975). The oldest source where *H. lupulus* is mentioned is from A.D. 768, when king Pipin, father of Charlemagne donated *Humlonarias cum integritate* to the monastery of St. Denis in Paris. However, *humlonariae*, or its French equivalent, has been regarded as a placename but without the economic implication of hops production. In the light of recent hop finds from the early Medieval period (see above), one is probably justified in surmising that this was a place where hops had at least been collected. Cultivation of *H. lupulus* at that time is unlikely since hop is not registered in the *Capitulaire de Villis* of Charlemagne (ca. A.D. 800). In this document, everything regarding the royal estates is regulated and all plants that had to be grown there are mentioned.

That *H. lupulus* was used in brewing is first documented in the statutes of the abbot Adalhardus from the monastery of Corvey on the river Weser in Germany (Fischer 1929; Wilson 1975). According to these statutes, hops were gathered at that time and this posed no problems since this plant occurred naturally in the floodplain forests along the river Weser.

Cultivation of *H. lupulus* appears for the first time in the documents of the Hochstift (monastery) Freising, Bavaria. Between A.D. 859 and 875, apart from orchards, fields and meadows, *humularia*, i.e. hop gardens, are mentioned in the sources. From the middle of the ninth century, so many duties on hops are known to have been levied on the tenants of monasteries in France and elsewhere, that intentional cultivation must be postulated.

During the early and high Medieval period, hops were the common beer additive outside the area in which *M. gale* was native. There was strong competition, however, in the frontier areas where both sorts of beer were available. Many regulations of this competition are set out in old documents. An important advantage of hop beer in contrast to *grut* beer was that it travelled better. Already in the thirteenth century, hop beer from Bremen was exported in large quantities mainly to the Netherlands and Flanders. In the following century Hamburg followed Bremen as a beer exporter. The former became the beer town of the Hanse or Hanseatic League. According to Bracker (1994), there were 457 hop breweries in Hamburg in the year 1369, with the product from 181 breweries being used exclusively for export to the Netherlands.

Hop beer from Lübeck, Wismar or Danzig, that was often referred to as German or Prussian beer, was also exported in large quantities to Denmark and Sweden from the thirteenth century onwards and particularly in the fourteenth century when it became common in these countries. The production of hop beer in these regions followed its export and so the cultivation of *H. lupulus* expanded to the north and west. Hops also became an important traded commodity and is reported as such in many written sources. Finds of hop fruitlets may even indicate import as has been shown for instance in the finds of the late Medieval brewery in Bergen in Norway (Krzywinsky and Soltved 1988).

From the fourteenth century onwards, *Myrica* beer was in decline everywhere in its north-west European home area except for Britain where this occurred later (Behre 1998; Kaiser 1995; van Vilsteren 1994). It was defended, however, as much as possible by its producers. Then rumours were spread that beer from sweet gale was harmful for drinkers. Excessive drinking was said to lead to blindness and even death. Consequently, the brewing of this beer was forbidden. In Mölln, Schleswig-Holstein, brewers who used sweet gale were being fined already in the sixteenth century, and in most parts of north-west Germany the production of *Myrica* beer was forbidden in the early eighteenth century (in 1723 within the electorate of Hannover). This resulted in the cessation of production of sweet gale beer. However, chemical investigations carried out in recent times have failed to find ingredients in *M. gale* that are poisonous.

At present, *H. lupulus* is the only permitted beer additive. Until the eighteenth century, hop was grown in many areas of central Europe, but nowadays its production has been concentrated in a few regions only, mainly in Bavaria and the Czech Republic.

Other plants reported for flavouring beer

In early times there were no or only few regulations for the production of beer, so many aromatic and other plant species were added to beer to improve its taste or for other purposes. The archaeobotanical record contains many of plants that might have been used in this way but, on the whole, there is no certain proof of their use. Written sources only are therefore employed as evidence for such use.

Table 1. Other plant species used for beer flavouring

<i>Anchusa officinalis</i> L.	<i>Phyllitis scolopendrium</i>
<i>Artemisia absinthium</i> L.	(L.) NEWM.
<i>Artemisia vulgaris</i> L.	<i>Picea abies</i> (L.) KARST.
<i>Asarum europaeum</i> L.	<i>Pimpinella anisum</i> L.
<i>Calluna vulgaris</i> (L.) HULL.	<i>Potentilla anserina</i> L.
<i>Citrus limon</i> BURM.	<i>Prunus avium</i> L.
<i>Cnicus benedictus</i> L.	<i>Prunus spinosa</i> L.
<i>Euphrasia rostkoviana</i> HAYNE	<i>Quercus</i> sp.
<i>Foeniculum vulgare</i> MILL.	<i>Rosmarinus officinalis</i> L.
<i>Fragaria vesca</i> L.	<i>Rubus fruticosus</i> L.
<i>Geum urbanum</i> L.	<i>Rubus idaeus</i> L.
<i>Hyssopus officinalis</i> L.	<i>Salvia officinalis</i> L.
<i>Inula helenium</i> L.	<i>Sambucus nigra</i> L.
<i>Juniperus communis</i> L.	<i>Sanicula europaea</i> L.
<i>Laurus nobilis</i> L.	<i>Stachys officinalis</i> (L.) TREV.
<i>Lavandula angustifolia</i> MILL.	<i>Szygium aromaticum</i> (L.) MERR.
<i>Majorana hortensis</i> MOENCH	et PERRY
<i>Melissa officinalis</i> L.	<i>Teucrium scordium</i> L.
<i>Mentha pulegium</i> L.	<i>Thymus serpyllum</i> L.
<i>Mentha spicata</i> L.	<i>Veronica officinalis</i> L.
<i>Origanum vulgare</i> L.	<i>Vinca minor</i> L.
	<i>Zingiber officinale</i> ROSCOE

Several ancient herbals and medical books, from all parts of Europe, refer to the use of certain plants in brewing. Some of these books go back to the Middle Ages,

but most date to the 1500s and onwards. Special textbooks were also written for brewers, for instance *Der vollkommene Bierbrauer* which was published anonymously in Frankfurt and Leipzig (1784). In this book, many species are mentioned which were added to beer only 200 years ago even though, in much of Germany, beer purity laws (*Reinheitsgebote*) were already in force.

These beer additives were used for several purposes. Most of them served as flavouring agents to give the beer a better taste. The list presented in Table 1, which is certainly not complete, gives a number of more important, mainly aromatic species, that have been used to flavour beer. They did not act as preservatives and they were generally used in addition to hops. Some of these species have been in use until this century, for instance *Ledum palustre* and *Juniperus communis* in Estonia (Leinbock 1932). In most cases, stems, roots, leaves and flowers that had been dried were used, rather than the fruits or seeds. The list also includes sweet fruits like cherries, sloes and raspberries which were put into the beer, and sometimes sugar or honey was also added. Resin or sulphur might be added to the list and also other materials.

Table 2. Plants used for medicinal beers (as far as not yet included in table 1)

<i>Acorus calamus</i> L.
<i>Carum carvi</i> L.
<i>Cinnamomum aromaticum</i> NEES
<i>Cinnamomum verum</i> PRESL.
<i>Elettaria cardamomum</i> (L.) MATON
<i>Galium odoratum</i> (L.) SCOP.
<i>Hepatica nobilis</i> MILL.
<i>Hypericum perforatum</i> L.
<i>Iris</i> sp.
<i>Ledum palustre</i> L.
<i>Meum athamanticum</i> JACQ.
<i>Myristica fragrans</i> VAN HOUTTE
<i>Petroselinum crispum</i> (Mill.) A.W. HILL
<i>Polypodium vulgare</i> L.

Other additives had medicinal properties but these included plants that, in fact, were poisonous. Various plants were used to prepare a variety of medicinal beers against plague and other diseases. Some of these species are already included in Table 1 while further species are listed in Table 2. Poisonous plants that were added (sources include Tabernaemontanus 1731) include *Lolium temulentum* L., *Anamirta cocculus*, *Hyoscyamus niger* L. and even *Atropa bella-donna* L., *Datura stramonium* L. and fly agaric, *Amanita muscaria* L. ex FR. Special beers with poisonous ingredients like the ones above were the so-called *berserker* beers which were given to the lansquenets to increase their rage against the enemy.

Outside of the hop growing regions, substitutes for hops were sometimes used. This was the case in Iceland, where *Achillea millefolium* L. was used instead of hops and was therefore called *valhumal* (field hop). This species was also used in parts of Sweden until the eighteenth century (Grässe 1872). Even in central Europe,

hops were sometimes substituted for by other plants such as the leaves of *Fraxinus excelsior* L. (Heilige Hildegard) or *Salix* spp. (Tabernaemontanus 1731).

The use of the many plants that have been mentioned here (and others, too!) resulted in a great variety of beers. All available cereals were also used in the production of malt and this has additional variety. Each town and sometimes each brewer had their own specific beer in contrast to the rather uniform modern beers, where only hops are allowed as an additive.

Conclusions

The sharp increase in archaeobotanical investigations, particularly relating to the Middle Ages, has resulted in a much better knowledge of the plants used as beer additives, including plants that had often been disregarded in earlier times. Since the first compilation regarding the use of *M. gale* and *H. lupulus*, i.e. the two most important species (Behre 1984), the number of sites with *M. gale* and *H. lupulus* has risen from 16 to 56 and 27 to 119, respectively. The revised site distribution maps (Figs. 3, 4, 6) may be regarded as providing more or less reliable evidence for the use in past times of these plants in brewing. Although every effort has been made to compile fully comprehensive lists, there are undoubtedly some omissions for which the author wishes to apologize in advance.

In reading and interpreting the site distribution maps, the following points should be borne in mind:

1. As nearly all records for these two species come from non-carbonized material, the distribution maps are inevitably influenced by the availability of waterlogged sites.
2. Medieval archaeology, which provides most of these finds, is a comparably young discipline and is not yet fully developed in all regions.
3. Archaeobotanists may not be available in particular excavations and/or regions to carry out the necessary identifications.

The three points listed above clearly have a bearing on the distribution maps in that regions with a good number of waterlogged sites, where several Medieval excavations have been carried out and where high standards of archaeobotanical investigation prevail, can be readily identified. In regions like the Netherlands, all three of these factors come together and result in maps with many sites.

Comparison of the archaeobotanical and written records for the use of *M. gale* suggests that the conclusions reached here are reliable. In most parts of its area of natural distribution, the remains of *M. gale* have been found in archaeological contexts from the early Middle Ages and sometimes in considerable quantities. The oldest documents, which are from the Netherlands and Belgium, also go back to this period and confirm the use of *M. gale* for brewing. Some early finds of macroremains go back to the Roman and even the pre-Roman (7 sites) Iron Age. They are confined to the region of the Rhine estuary in the northern Netherlands and all are from

clear archaeological contexts. This suggests that sweet gale was already used here at a very early date, probably for making beer. Written sources, of course, are not available for this period.

The history of *H. lupulus* as beer additive is slightly different from that of *M. gale*. There are only a few sites with single hop fruitlets from pre-Medieval times so that there is little or no evidence for use at this time. Roman writers refer repeatedly to Germanic and Celtic beer, and they mention cereals but never hops. During the early Middle Ages, and in a few cases already before A.D. 800, there is a very steep increase in the number of sites with hops as well as in the quantity of remains. This indicates undoubtedly the utilization of this species. The archaeobotanical record fully corresponds with the written sources that document the use of *H. lupulus* for brewing since the early ninth century. The oldest documents indicate that hops were collected in the wild. The cultivation of hop is first recorded for A.D. 859 in Bavaria.

Until the thirteenth century, sweet gale was the prevailing beer additive inside its natural distribution area and hops dominated outside this region. Later, strong competition developed and hop beer and later hop cultivation extended into the home area of sweet gale. The use of *M. gale* for beer ended in the middle of the mid-eighteenth century.

There are many other herbs of secondary importance which were used to improve the taste of beer or to prepare medicinal beers. This can be gathered from written sources only, but these are plentiful. Until the eighteenth century, there was a great variety of beers which since then has been reduced to the very uniform beer culture of nowadays in which hop is the only accepted additive.

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References

- Aalto M (1982) Archaeobotanical studies at Katajamäki, Isokylä, Salo, south-west Finland. *Pact* 7: 137-147
- Aalto M (1994) [The vegetation of the Turku castle]. *Turun Maakuntamuseo Raportteja* 16: 21-38
- Alsleben A (1991) Archäobotanische Untersuchungen in der Hansestadt Lübeck: Landschaftsentwicklung im städtischen Umfeld und Nahrungswirtschaft während des Mittelalters bis in die frühe Neuzeit. *Offa* 48: 329-362
- Alsleben A (1997) Nutzpflanzen aus dem mittelalterlichen Wolin: zwei ausgewählte Gruppen: Getreide und Lein. *Offa* 52 (1995): 185-217
- Averdieck FR (1990) Botanische Bearbeitung von Proben der Grabungsplätze Heiligen-Geist-Hospital und Königstraße in Lübeck. *Offa* 46: 307-332
- Bakels CC (1980) Een Sittardse beerput en mestvaalt: plantenresten uit het laat-middeleeuwse gasthuis "aan de bruggen bij de markt". *Archeol Limburg* 9: 2-3

- Baudais-Lundstrom K (1978) Plant remains from a Swiss neolithic lakeshore site: Brise-Lames, Auvernier. *Ber Dtsch Bot Ges* 91: 67-83
- Behre KE (1976) Die Pflanzenreste aus der frühgeschichtlichen Wurt Elisenhof. *Studien zur Küstenarchäologie Schleswig-Holsteins, Reihe A: Elisenhof 2*, Frankfurt/M.
- Behre KE (1981) Pflanzenreste der Zeit um 1400 n. Chr. aus dem Lüneburger St.-Michaelis-Kloster. *Nachr Nieders Urgesch* 50: 321-327
- Behre KE (1983) Ernährung und Umwelt der wikingerzeitlichen Siedlung Haithabu: die Ergebnisse der Untersuchungen der Pflanzenreste. *Ausgr Haithabu 8*, Neumünster
- Behre KE (1984) Zur Geschichte der Bierwürzen nach Frucht-funden und schriftlichen Quellen. In: Zeist W van, Casparie WA (eds) *Plants and ancient man. Studies in palaeoethnobotany*, Rotterdam, pp 115-122
- Behre KE (1986) Ackerbau, Vegetation und Umwelt im Bereich früh- und hochmittelalterlicher Siedlungen im Flußmarschgebiet der unteren Ems. *Probl Küstenforsch südl Nordseegeb* 16: 99-125
- Behre KE (1991) Die ersten Funde von Nahrungspflanzen aus dem Mittelalter Bremens. *Bremisches Jahrb* 70: 207-227
- Behre KE (1998) Zur Geschichte des Bieres und der Bierwürzen in Mitteleuropa. *Archäol Mitt Nordwestdeutschland* 20: 49-88
- Behre K-E, Kučan D (1994) Die Geschichte der Kulturlandschaft und des Ackerbaus in der Siedlungskammer Flögeln, Niedersachsen, seit der Jungsteinzeit. *Probl Küstenforsch südl Nordseegeb* 21: 1-227
- Bencard M, Lange J (1972) Botaniske resultater af en udgravning i Ribe. *Mark Monte* 8: 27-37
- Botanisch Laboratorium (1982) Project urbanisatie in het Nederlandse rivierengebied tijdens de Middeleeuwen: Deventer. *ROB jaarverslag* 1982: 92-94
- Bracker J (1994) Hopbier uit Hamburg. In: Kistenaker RE, Vilsteren VT van (eds) *Bier! Geschiedenis van een volksdrank. De Bataafsche Leeuw*, Amsterdam 1994: 28-33
- Brink LM van den (1984) Dorstige Harthof en omgeving. *Maandblad Oud Utrecht* 8/9: 108-113
- Brink LM van den (1988) Zaden en pollen uit een 16e eeuwse beerput uit de Postelstraat. In: Boekwijt HW en Janssen HL (eds) *Kroniek van bouwhistorisch en archeologisch onderzoek 's-Hertogenbosch*. pp 113-124, 's-Hertogenbosch
- Brink W van den (1989) Zaden en stuifmeel uit een put in "Den Prince van Luyck". In: Heymans H (ed) *Van put naar kluis*, pp 266-276, Maaseik
- Brinkkemper O (1993a) Botanische macroresten van Rockanje 08-54, een vindplaats uit de Late IJzertijd. *Intern Rapport I.P.L.*
- Brinkkemper O (1993b) Wetland farming in the area to the south of the Meuse estuary during the Iron Age and Roman Period: an environmental and palaeoeconomic reconstruction. *Analecta Praehist Leidensia* 24, (1991)
- Brinkkemper O (1997a) Rotterdam en omgeving van de IJzertijd tot en met de Late Middeleeuwen; een milieu-reconstructie op basis van onderzoek van plantenresten. *Boorbalans* 3: 45-71
- Brinkkemper O (1997b) Botanische gegevens. In: Carmiggelt A (ed) *Laat- en postmiddeleeuwse bewoningssporen aan de Hoogstraat te Rotterdam*, Boorbalans 3: 252-258 and pp 276-278
- Brinkkemper O, Vermeeren C (1994) Mediterrane rijst en oosterse kruidnagels: botanisch onderzoek aan een beerkelder uit Kampen (1575-1650). *Biaxiaal* 3
- Brombacher Chr, Jacomet St, Kühn M (1997) Mittelalterliche Kulturpflanzen aus der Schweiz und Liechtenstein: eine Übersicht der archäobotanischen Nachweise. In: Boe G de, Verhaeghe F (eds) *Medieval Europe Brugge 1997. Conference volume* 90: 95-111
- Buurman J (1976) Botanisch onderzoek van een Middeleeuwse stadsgracht te Enschede. *t Inschien* 8: 59-61
- Cappers RTJ (1995) Botanical macro-remains of vascular plants of the Heveskesklooster terp (the Netherlands) as tools to characterize the past environment. *Palaeohistoria* 35/36: 107-167
- Čulíková V (1987) Ein interessanter Fund von pflanzlichen Makro-resten aus dem mittelalterlichen Prag. *Archeol Rozhledy* 49: 445-452
- Čulíková V (1995) Rekonstruktion der synanthropen Vegetation des mittelalterlichen Most. *Památky archeol* 86(1): 83-131
- Der vollkommene Bierbrauer 1784, Wendlern, Frankfurt und Leipzig
- Esser E, Gehasse EF (1995) Onderzoek van huisafval: het organisch materiaal. In: Bitter P (ed) *Geworteld in de bodem: archeologisch en historisch onderzoek van een pottenbakkerij bij de Wortelsteeg in Alkmaar*, pp 77-87
- Fischer H (1929) *Mittelalterliche Pflanzenkunde*. München. Nachdruck Olms 1967, Hildesheim
- Gluza I (1970) Les restes végétaux du haut Moyen Age provenant de la tranchée V située à la Skarpa à Cracovie. *Materiały archeologiczne* 11: 267-269
- Godwin H, Bachem K (1961) Plant material. *York. Archaeol J* 116: 109-113
- Gräse IHT (1872) *Bierstudien*. Dresden
- Green FJ (1983) The plant remains. In: Millett M (ed) *The history, architecture and archaeology of Johnson's Corner, Alton. Proc. Hampsh. Field Club Archaeol Soc* 39: 1-105
- Greig JRA (1993) Römische und mittelalterliche Pflanzenreste aus Lincoln, England. *Archaeo-Physika* 13: 139-148
- Griffin K (1977) Plant remains from medieval Oslo. *Universitetets Oldsaksamling Årbok* 1975/1976: 151-163
- Griffin K (1979) Plant remains from "Oslogate 7". *De arkeologiske utgravninger i gamlebyen, Oslo* 2: 124-133
- Griffin K (1981) Planterester. *Norw Antiq Bull* 4: 272-282
- Griffin K (1982) Utgravningen på Helgeandsholmen: analyser av jordprover från sju båtar. *Rapport RAÄ* 1982, 2: 1-25
- Griffin K, Sandvik PU (1989) Frukter, frø og andre makrofossiler: funksjoner og aktiviteter belyst gjennom analyser av jordprover. *Fortiden Trondheim Bygrunn Folkebibliotekstomten Medd.* 19: 1-108
- Haaster H van (1985) Schagen-Muggenburg: een bijdrage aan de reconstructie van milieu en agrarische activiteiten rond een nederzetting uit de 2e/3e eeuw. Unveröff. Examensarbeit, University of Amsterdam
- Haaster H van (1997) Plantaardige en dierlijke resten uit de Middeleeuwen: de resultaten van het oecologisch onderzoek op het Sint Janskerkhof. In: Boekwijt HW, Janssen HL (eds) *Bouwen en wonen in de schaduw van de Sint Jan. Bouw-historisch en archeologisch onderzoek 's-Hertogenbosch* 2, 's-Hertogenbosch
- Hänninen K, Man R de (1994) Stuifmeel en zaden, In: Krauwer M, Snieder F (eds) *Nering en vermaak: de opgraving van een veertiende eeuwse markt in Amersfoort, Utrecht*, pp 163-168
- Hall AR (1992) The last teasel factory in Britain, and some observations on teasel (*Dipsacus fullonum* L. and *D. sativus* (L.) Honckeney) remains from archaeological deposits. *Circaea* 9(1): 9-13
- Hall AR, Kenward HK (1990) Environmental evidence from the Colonia: General Accident and Rougier Street. *Archaeol. York* 14(6): 289-434
- Hansson AM (1997) Finds of hops, *Humulus lupulus* L., in the black earth at Birka, Sweden. *Theses and Papers in Archaeology* B5: 129-137
- Hegi G (1957) *Illustrierte Flora von Mitteleuropa* III/1, 2nd edn., Hanser, München
- Heinrich D, Schulz F, Wiethold J (1994) Archäobotanische und archäozoologische Funde aus den mittelalterlichen Wehrgräben von Kiel. *Archäol Nachr Schleswig-Holstein* 4/5 (1993/94): 70-102

- Helck W (1971) Das Bier im Alten Ägypten. Berlin
- Hellwig M (1990) Paläoethnobotanische Untersuchungen an mittelalterlichen und frühneuzeitlichen Pflanzenresten aus Braunschweig. *Diss Bot* 156: 1-210
- Hellwig M (1997) Plant remains from two cesspits (15th and 16th century) and a pond (13th century) from Göttingen, southern Lower Saxony, Germany. *Veget Hist Archaeobot* 6: 105-116
- Hellwig M, Kuprat B (1991) Paläoethnobotanische Befunde aus einem frühneuzeitlichen Brunnen in Northheim. *Northheimer Jahrb* 56: 96-107
- Hingh A de, Bakels C (1996) Palaeobotanical evidence for social differences? The example of the early medieval domain of Serris-Les-Ruelles, France. *Veget Hist Archaeobot* 5: 117-120
- Hjelmqvist H (1991) Några trädgårdsväxter från Lunds medeltid. *Svensk Bot.Tidskr* 85: 225-248
- Hjelmqvist H (1992) Some economic plants from the prehistoric and medieval periods in southern Scania. In: Larsson L, Callmer J, Stjernqvist B (eds) *The archaeology of the cultural landscape. Acta Archaeol Lundensia Ser in 4^o, 19*: 359-367
- Hjelmqvist H (1995) Cayennepeppar från Lunds medeltid. *Svensk Bot Tidskr* 89: 193-201
- Hollevoet Y, Cooremans B, Desender K, Eryvynck A (1993) Een Karolingische vlechtwerkwaterput uit Zerkegem (gem. Jabbeke, prov. West-Vlaanderen): culturele en ecologische archaeologica. *Archeol Vlaanderen* 3: 243-254
- Jensen HA (1979) Seeds and other diaspores in medieval layers from Svendborg. *The Archaeology of Svendborg, Denmark*, 2, Odense University Press, Odense.
- Jensen HA (1986) Seeds and other diaspores in soil samples from Danish town and monastery excavations, dated 700-1536 AD. *Biol Skr* 26: 1-107
- Jessen K (1954) Plantefund fra vikingetiden i Danmark. *Bot Tidsskr* 50: 125-139
- Jørgensen G (1986) Medieval plant remains from the settlements in Møllergade 6. *The Archaeology of Svendborg, Denmark*, 4: 45-84
- Kaiser H (1995) Der große Durst. Materialien und Studien zur Alltagsgeschichte und Volkskultur Niedersachsens 23, Museumsdorf Cloppenburg, 1-254
- Kenward HK, Hall AR (1995) Biological evidence from Anglo-Scandinavian deposits at 16-22 Coppergate. *Archaeol York* 14(7): 435-797
- Klichowska M (1972) Vascular plants in archaeological excavations of North-Western Poland from the Neolithic to the Early Middle Ages. *Poznańskie Towarzystwo Przyjaciół Nauk/Prace Komisji Biologicznej* 35, (2): 31-73
- Klichowska M (1975) Subfossile Pflanzen aus der Niederungsburg zu Bnin, Kr. Śrem. *UAM-Archeologia* 7: 185-198
- Knörzer KH (1968) Ein Teilergebnis der Untersuchung pflanzlicher Großreste bei der Ausgrabung an der Niederungsburg bei Haus Meer, Gemeinde Büderich. *Rheinische Ausgr* 1: 87-100
- Knörzer KH (1979) Spätmittelalterliche Pflanzenreste aus der Burg Brüggen, Kr. Viersen. *Bonner Jahrb* 179: 595-611
- Knörzer KH (1981) Römerzeitliche Pflanzenfunde aus Xanten. *Archaeo-Physika* 11: 176 pp
- Knörzer KH (1984) Aussagemöglichkeiten von paläoethnobotanischen Latrinenuntersuchungen. In: Zeist W van, Casparie WA (eds) *Plants and ancient man: studies in palaeoethnobotany*. Balkema, Rotterdam, pp 331-338
- Knörzer KH (1988) Pflanzenreste aus einer spätmittelalterlichen Schachtlatrine in Neuss. *Neusser Jahrb*, pp 42-51
- Knörzer KH (1991a) Pflanzliche Großreste von der Burg Wachtendonk. *Bonner Jahrb* 191: 503-511
- Knörzer KH (1991b) Reis, Buchweizen und Johannisbeere - mittelalterliche Pflanzenfunde aus einer Grube an der Agrippastrasse in Köln. *Kölner Jahrb Vor- u. Frühgesch* 24: 495-508
- Knörzer KH (1992) Vorbericht über paläo-ethnobotanische Untersuchungen in Duisburg. In: Krause G (ed) *Stadtarchäologie in Duisburg 1980-1990*. Duisburger Forsch 38: 223-236
- Knörzer KH (1994) Mittelalterliche und neuzeitliche Pflanzenreste aus der Stadt Siegburg. *Heimatbl Rhein-Sieg-Kreis* 62: 153-175
- Knörzer KH (1996) Pflanzentransport im Rhein zur Römerzeit, im Mittelalter und heute. *Decheniana* 149: 81-123
- Knörzer KH, Krull HP, Wirth S (1995) Der große Markt zu Wesel: Leder-, Pflanzen- und Knochenfunde aus der Grabung 1987/88. *Bonner Jahrb* 195: 371-423
- Knörzer KH, Müller G (1968) Mittelalterliche Fäkalien-Faßgrube mit Pflanzenresten aus Neuß. *Rheinische Ausgr* 1: 131-170
- Kooistra LI, Hessing WAM (1988) Ein frühmittelalterlicher Brunnen mit einer exotischen Frucht aus Houten. *Ber Rijksdienst Oudheidkundig Bodemonderzoek* 38: 207-228
- Kosina R (1977) Die Speicher von Wrocław aus dem XI. Jahrhundert: ein Beitrag zu Forschungen über Wirtschaft in diesem Raum. *Kwartalnik historii kultury materialnej* 2/77: 257-267
- Kosina R (1978) The cultivated and wild plants from the XIth century granaries on the cathedral-island in Wrocław. *Ber Dtsch Bot Ges* 91: 121-127
- Kosina R (1995) Botanical synopsis of medieval Wrocław. In: Kroll H, Pasternak R (eds) *Res archaeobotanicae - 9th symposium IWGP, Kiel 1992*. Oetger-Voges-Verlag, Kiel, pp 101-116
- Kreuz A (1993) Nutz- und Wildpflanzenfunde (Oberursel-Bommersheim). *Germania* 71: 502-509
- Kroll H (1991) Kultur- und Sammelpflanzen. In: Müller-Wille M (ed) *Starigard/Oldenburg: ein slawischer Herrschersitz des frühen Mittelalters in Ostholstein*, Wachholtz, Neumünster, pp 307-314
- Krzywinsky K, Soltvedt EC (1988) A medieval brewery (1200-1450) at Bryggen, Bergen. *Bryggen Papers/suppl ser* 3: 1-68
- Kučan D (1998) Zur Ernährungsgeschichte des Spätmittelalters und der frühen Neuzeit in Oldenburg anhand der botanischen Untersuchungen der Altstadtgrabungen. *Probl Küstenforsch südl Nordseegeb* 25
- Kühn F (1991) Ein Fund von Samen aus dem Mittelalter aus Iglau mit besonderer Hinsicht auf Steinkerne von *Prunus*. *Vlastiv dný sborník vysoiny* 10: 17-36
- Küster H (1988) Spätmittelalterliche Pflanzenreste aus einem Brunnen von Deggendorf (Niederbayern). In: Schmotz K (ed) *Vorträge des 6. Niederbayerischen Archäologentages, Deggendorf A 6/1*: 175-199
- Küster H (1989) Mittelalterliche Pflanzenreste aus Konstanz am Bodensee. *Diss Bot* 133: 201-216
- Küster H (1992) Mittelalterliche Pflanzenreste aus Eichstätt: botanische Analysen ermöglichen Einblicke ins tägliche Leben. In: Rieder KH, Tillmann A (eds) *Eichstätt: 10 Jahre Stadtkernarchäologie: Zwischenbilanz einer Chance*, Kipfenberg, pp 101-112
- Kuijper WJ (1982) Plantenresten uit de Rijnafzetting en de bewoningslagen onder de Breestraat te Leiden. *Bodemonderzoek Leiden (jaarverslag 1981)*
- Kuijper WJ (1985) Planten- en dierenresten in laat-middeleeuwse Beerputten op het terrein van het St. Agnietenklooster in Leiden. *Bodemonderzoek Leiden (jaarverslag 1984)* 7: 131-142
- Lange AG (1990) De Horden near Wijk bij Duurstede: plant remains from a native settlement at the Roman frontier: a numerical approach. *Nederlandse Oudheden 13/Kromme Rijn projekt 3*, Amersfoort
- Lange E (1987) Mittelalterliche Pflanzenfunde aus Magdeburg. *Z Archäol* 21: 243-256
- Lange E (1993) Zum Stand und zu einigen Ergebnissen paläo-ethnobotanischer Arbeiten in der Niederlausitz. *Biol Stud Luckau* 22: 14-26
- Latałowa M, Badura M (1996) XIX. Szczatki roślinne. In: Rebkowski M (ed) *Archeologia średniowiecznego Kołobrzegu: 1. Badania przy zł. Tatuszowej 9-13*: 385-418, Kołobrzeg
- Leinbock F (1932) Die materielle Kultur der Esten, Akademilne Koop., Tartu

- Lempiäinen T (1985) Plant remains from the ancient lake Mätäjärvi in Turku, SW Finland. *Iskos* 5: 258-271
- Lempiäinen T (1988) Pflanzliche Makroreste von dem Innenhof des Qwenselschen Anwesens in Turku, SW Finnland, vom 17.-19. Jh. *Ann Bot Fenn* 25: 47-54
- Lempiäinen T (1992) Pflanzliche Makroreste von der wikingerzeitlichen – frühmittelalterlichen Siedlung Varikkoniemi in Hämeenlinna. *Eripainos Suomen Museo, Vammala*, pp 109-128
- Lempiäinen T (1994) [Plant remains from the castle Kuusisto]. *Museoviraston Rakennushistorian Osaston Raportteja* 8: 80-99
- Lempiäinen T (1995) Macrofossil finds from the medieval Turku, SW Finland. In: Kroll H, Pasternak R (eds) *Res archaeobotanicae - 9th symposium IWGP, Kiel 1992*. Oetger-Voges-Verlag, Kiel, pp 149-164
- Luijten H (1992) Zaden en vruchten: overblijfselen van het plantaardige voedsel en de begroeiing van de grachten. In: Arts N (ed) *Het Kasteel van Eindhoven: archeologie, ecologie en geschiedenis van een heerlijke woning 1420-1676*, Eindhoven, pp 237-244
- Luijten H (1994) Gebruiksplanten en de natuurlijke vegetatie van het Heuvelterrein. In: Arts N (ed) *Sporen onder de Kempische stad*, Eindhoven, pp 302-312
- Lynch A, Paap N (1982) Untersuchungen an botanischen Funden aus der Lübecker Innenstadt (ein Vorbericht). *Lübecker Schr Archäol Kulturgesch* 6: 339-360
- Lynch A, Paap N (1986) Botanische Untersuchungen zur Grabung an der Untertrave 97 in Lübeck: ein Beitrag zu den naturräumlichen Voraussetzungen mittelalterlicher Siedlungsgeschichte. *Lübecker Schr Archäol Kulturgesch* 12: 15-25
- Maier U (1983) Nahrungspflanzen des späten Mittelalters aus Heidelberg und Ladenburg nach Bodenfunden aus einer Fäkaliengrube und einem Brunnen des 15./16. Jahrhunderts. *Forsch Ber Archäol Baden-Württ* 8: 139-183
- Moffett L (1992) Fruits, vegetables, herbs and other plants from the latrine at Dudley Castle in central England, used by the Royalist garrison during the Civil war. *Rev Palaeobot Palynol* 73: 271-286
- Moor A (1991) Die frühere Kost der estnischen Bauern. 2 vol. Valgus, Tallinn
- Murphy P (1988) Plant macrofossils (excluding wood and mosses). In: Ayers B (ed) *Excavations at St. Martin-at-Palace Plain, Norwich*. *East Anglian Archaeol* 37: 118-125
- Murphy P (1994) Plant macrofossils (excluding wood and mosses). *East Anglian Archaeol* 68: 54-58
- Murphy K, Nye S, Greig J, Phipps J, Evans J, Bell MG, Jones AKG, Williams MK (1989) The analyses of a cesspitt fill from the Tudor merchant's house, Tenby, Dyfed. *Bull Board Celtic Stud* 36: 246-262
- Opravil E (1969) Rostlinné nálezy z archeologického výzkumu středověké Opavy prováděného v roce 1967. *Časopis Slezského muzea/ser 1*, 18: 175-182
- Opravil E (1979) Pflanzenreste aus Mohelnice (Neolithikum, Bronzezeit). *Časopis Slezského Muzea Opava/ A* 28: 1-13 and 97-109
- Opravil E (1983) Die Talaue in der Burgwallzeit (Tschechoslowakei: Morava- und Odragebiet). *Studie Archeologického ústavu Československé Akademie věd v Brně* 11(2): 1-77
- Opravil E (1984) Pflanzenreste aus dem Hof des Kaufmannshauses in Olomouc. *Archeol Rozhledy* 36: 194-202
- Opravil E (1985a) Pflanzenreste aus der Abfallgrube in Tábor Nr. 6. *Archeol Rozhledy* 37: 186-194
- Opravil E (1985b) Ergebnisse archäobotanischer Analysen aus dem historischen Stadtkern von Uherské Hradiště. *Přehled výzkumu* 1985: 45-82
- Opravil E (1986) Archäobotanische Funde aus dem Areal Jaktarer Tor in Opava (eh. Hotel "Koruna"). *Časopis Slezského Muzea Opava/A* 35: 227-253
- Opravil E (1990a) Die Vegetation in der jüngeren Burgwallzeit in Přerov. *Časopis Slezského Muzea Opava/A* 39: 1-22
- Opravil E (1990b) Post-mediaeval archaeobotanical finds from the town of Olomouc. *Stud Postmed Archaeol* 1: pp 231-248
- Opravil E (1990c) Archäobotanische Funde aus der Kolářská-Gasse in Opava. *Archäol hist* 15: 491-509
- Opravil E (1993) Rostliny z středověku Uherského Brodu - soukenická ulice a Lidový dům (okr. Uherské Hradiště). *Přehled výzkumu* 1989: 135-143
- Opravil E (1994) Synanthrope Vegetation der Stadt Olomouc aus dem Mittelalter und aus dem Anfang der Neuzeit. *Zpr. Čes. Bot. Společ.* 29(11): 15-36
- Opravil E (1995) Sortiment sběrného hospodářství v archeologických nálezech z historického jádra města Opavy. *Sborník památkového ústavu v ostrave* 1995: 37-41
- Opravil E (1996) Archaeobotanical finds from the historical kernel from the research season 1993-1994. *Časopis Slezského Muzea Opava/A* 45: 1-15
- Paap NA (1983) Economic plants in Amsterdam: qualitative and quantitative analysis. In: Jones M (ed) *Integrating the subsistence economy: symposia of the Association for Environmental Archaeology*. BAR International Series 181, Oxford, pp 315-325
- Paap NA (1984) Botanische Analysen in Lübeck: eine Zwischenbilanz. *Lübecker Schr Archäol Kulturgesch* 8: 49-55
- Pals JP (1972a) Zadenanalyse stadskernonderzoek Dordrecht. Unveröff. Manuskript
- Pals JP (1972b) Plantaardigheden onder Damrak 49. Unveröff. Manuskript
- Pals JP (1980) Zaden en vruchten. In: Feenstra EJ, *Archeologische waarnemingen in Harderwijk*. *Westerheem* 29(4): 299-300
- Pals JP (1986) Plantenresten uit een vroeg-middeleeuwse kreek te Leiderdorp. *Westerheem* 35: 236-241
- Pals JP (1988) Akkerbouw in het middeleeuwse Dommelen. *Phyto-archeologische studies*, pp 111-135. Doctoral thesis, University of Amsterdam
- Pals-van Blerk I (1986) Zaden en vruchten. In: Stoecker H (ed) *Graven naar het kasteel van Tilburg*. *Waalre*
- Pasternak R (1991) Hafer aus dem mittelalterlichen Schleswig. *Offa* 48: 363-380
- Robinson DE (1994) Plants and vikings: everyday life in Viking Age Denmark. *Bot J Scotland* 46(4): 542-551
- Robinson DE, Boldsen I (1998) Arkæobotanisk analyse. *Viborg Sønder 1000-1300*. *Jysk Arkæol Selskabs Skr* 34: 309-323
- Rösch M (1992) Zum Fortgang der Untersuchungen im frühmittelalterlichen Gräberfeld, Adelshof und Hofgrablege bei Lauchheim, Ostalbkreis. *Archäol Ausgr Baden-Württ* 1992: 231-243
- Rösch M, Fischer E (1995) Zum Fortgang der Untersuchungen in Schwäbisch Hall. *Archäol Ausgr Baden-Württ* 1994: 323-325
- Rösch M, Gross U (1994) Hochmittelalterliche Nahrungspflanzenvorräte aus Gerlingen, Kreis Ludwigsburg. *Fundber Baden-Württ* 19: 711-759
- Rösch M, Karg S, Sillmann M (1994) Vierhundert Jahre gelagert: Pflanzenreste in Decken und Wänden: botanische Dokumente zu Ernährung, Landwirtschaft und Landschaft aus der Langen Straße 49. *Kataloge des Hällisch-Fränkischen Museums Schwäbisch Hall* 8: 475-491
- Rostrup O (1906) Frø resp. frugter og andre plantelevninger m. m. In: Rosenkjaer HN (ed) *Fra det underjordiske København*, pp 91-143
- Ruas MP (1988) Alimentation végétale, pratiques agricoles et environnement, du VIIIe au Xe siècle (Villiers-le-Sec et Baillet-en-France). In: *Réunion des Musées Nationaux (ed) Un village au temps de Charlemagne*. Réunion des musées nationaux, Paris, pp 203-231

- Sandvik PU (1995) The Archbishop's Palace, Trondheim, Norway – results from the archaeobotanical investigations in 1991. In: Kroll H, Pasternak R (eds) *Res archaeobotanicae* - 9th symposium IWGP, Kiel 1992. Oetger-Voges-Verlag, Kiel, pp 267-279
- Schumann M (1990) Untersuchungen zur Kulturlandschaftsentwicklung Berlins: fossile Pflanzenreste aus frühgeschichtlichen und mittelalterlichen Siedlungsplätzen. Berlin-Forschung, FU, Berlin
- Sillasoo Ü (1996) Daily food and meal traditions in late medieval Tartu, Estonia (14th and 15th centuries). M.A. Thesis in Medieval Studies, Central European University Budapest, Budapest
- Stika HP (1996) Keltisches Bier aus Hochdorf. *Schr Keltenmus Hochdorf/Enz* 1: 64-75
- Suchá R, Koar P (1996) Ergebnisse der archäobotanischen makrorestlichen Analyse der mittelalterlichen Wasserleitung in Prachatitz. *Zlatá stezka* 3: 189-203
- Tabernaemontanus JT (1731) *Kräuterbuch*. Basel. Nachdruck Kölbl 1970, München
- Tallantire PA (1979) Late Viking and early Medieval plant material from Trondheim: a problem in interpretation. *Archaeo-Physika* 8: 295-301
- Thomas I, Greig J (1992) Plant remains. In: Rahtz P, Meeson R (eds) *An Anglo-Saxon watermill at Tamworth*. CBA Res Rep 83: 92-100
- Veen M van der (1996) Plant remains. In: Jackson RPJ, Potter TW (eds) *Excavations at Stonea, Cambridgeshire 1980-85*. British Museum Press, London, pp 613-639
- Vermeeren C (1990) Botanisch onderzoek van middeleeuwse beerputten uit Kampen. In: Clevis H, Smit M (eds) *Verscholen in vuil: archeologische vondsten uit Kampen 1375-1925*. Stichting Archeologie IJssel, Kampen, pp 139-161
- Vijverberg P (1993) Zaden en pitten. In: Koopmans B, Tuinstra B, Westenbroek B (eds) *Het huis te Palenstein in Zegwaart: historisch en archeologisch onderzoek naar een kasteel in het veen*, pp 105-108
- Vilsteren VT van (1983) Aan de vruchten kent men de... Voorst: botanisch onderzoek bij de opgraving. *Vereeniging tot beoefening van Overijsselsch regt en geschiedenis/werken* 36: 133-143
- Vilsteren VT van (1994) In de beginne... De oorsprong en techniek van het brouwen tot de 14de eeuw. In: Kistenaker RE, Vilsteren VT van (eds) *Bier! Geschiedenis van een volksdrank*. De Bataafsche Leeuw, Amsterdam, pp 7-19
- Vuorela I, Lempiaäinen T (1993) Palynological and palaeobotanical investigations in the area of the post-medieval Helsinki old town. *Veget Hist Archaeobot* 2: 101-123
- Wasylikowa K (1979) Plant remains from early and late medieval time found on the Wawel Hill in Cracow. *Acta Palaeobot* 19: 115-200
- Wasylikowa K, Zemanek A (1995) Plant and man in the Medieval Cracow. *Materiały Archeol* 28: 37-47
- Wiethold J (1993) Botanische Großreste des hohen und späten Mittelalters der Grabung Ulm, Donaustadt: erste Ergebnisse zur Ernährung und Umwelt im hoch- und spätmittelalterlichen Ulm. *Fundber Baden-Württ* 18: 493-599
- Wiethold J (1995a) Ein Blick auf den Speisezettel Lüneburger Patrizierfamilien im 16. und 17. Jahrhundert: archäobotanische Untersuchungen in Lüneburg. *Aufrisse (Mitt Arbeitskr Lüneburger Altstadt)* 11: 65-74
- Wiethold J (1995b) Plant remains from town-moats and cesspits of medieval and post-medieval Kiel (Schleswig-Holstein, Germany). In: Kroll H, Pasternak R (eds) *Res archaeobotanicae* - 9th symposium IWGP, Kiel 1992. Oetger-Voges-Verlag, Kiel, pp 359-384
- Wiethold J (1996) "Wyltu maken en gud moes van brambeeren..." – archäobotanische Untersuchungen in Lüneburg: kleine Pflanzenreste als Forschungsobjekt. In: Andraschko F, Lamschus H, Lamschus C, Ring E (eds) *Ton, Steine, Scherben: ausgegraben und erforscht in der Lüneburger Altstadt*. Deutsches Salz-mus Lüneburg, pp 112-125
- Wiethold J, Schulz F (1991) Pflanzliche Großreste aus einer Kloake des 16. Jahrhunderts aus der Grabung Kiel-Klosterkirchhof/Hassstrasse (LA 23). *Archäol Nachr Schleswig-Holstein* 2: 44-77
- Willerding U (1983) Paläo-Ethnobotanik und Ökologie. *Verhandlungen Gesellschaft für Ökologie* 11: 489-503
- Willerding U (1986) Paläo-ethnobotanische Befunde zum Mittelalter in Hörter/Weser. *Neue Ausgr Forsch Nieders* 17: 319-346
- Willerding U (1987) Landnutzung und Ernährung. In: Denecke D, Kühn HM (eds) *Von den Anfängen bis zum Ende des Dreißigjährigen Krieges*. Göttingen 1: 437-464, Göttingen
- Willerding U (1990) Botanische Befunde aus der Kloake Domhof 15/16 in Hildesheim. *Küche, Keller, Kemenate: Alltagsleben auf dem Domhof um 1600*, pp 96-106, Hildesheim
- Wilson DG (1975) Plant remains from the Graveney boat and the early history of *Humulus lupulus* L. in W. Europe. *New Phytol* 75: 627-648
- Wilson DG (1978) Early plants in the Nene Valley: an interim report. *Durobrivae* 6: 16-19
- Wolf G (1991) Die pflanzlichen Makroreste aus der mittelalterlichen Kloake Stummrig Str. 20 in Hörter/Westfalen. *Ausgr Funde Westfalen-Lippe* B 6: 507-523
- Zeist W van (1974) Palaeobotanical studies of settlement sites in the coastal area of the Netherlands. *Palaeohistoria* 16: 223-371
- Zeist W van (1983) Plant remains from Iron Age Noordbarge, province of Drenthe, the Netherlands. *Palaeohistoria* 23 (1981): 169-193
- Zeist W van (1991) Economic aspects. In: Zeist W van, Wasylikowa K, Behre KE (eds) *Progress in Old World palaeoethnobotany*. Balkema, Rotterdam, pp 109-130
- Zeist W van (1992) Cultuurgewassen en wilde planten. In: Broekhuizen PH, van Gangelen H, Helfrich K, Kortekaas GLGA, Alma RH, Waterbolk HT (eds) *Van boeren erf tot bibliotheek: historisch, bouwhistorisch en archeologisch onderzoek van het voormalig Wolters-Noordhoff-Complex te Groningen*. Stichting Monument & Materiaal, Groningen, pp 525-535
- Zeist W van (1993) Subfossiele zaden en vruchten: een onderzoek naar vegetatie, akkerbouw en voeding. In: Lenting JJ, Gangelen H van, Westing H van (eds) *Schans op de grens: Bourtanger bodemvondsten 1580-1850*. Stichting Vesting Bourtange, Sellinger, pp 597-619
- Zeist W van, Cappers R, Neef R, During H (1987) A palaeobotanical investigation of medieval occupation deposits in Leeuwarden, the Netherlands. *Proc K Nederlandse Akad Wetensch* B 90(4): 371-426
- Zeist W van, Roller GJ de, Palfenier-Vegter RM, Harsema OH, During H (1986) Plant remains from medieval sites in Drenthe, the Netherlands. *Helinium* 26: 226-274
- Zeist W van, Palfenier-Vegter RM (1995) Medieval plant remains from Peel, the Netherlands. *Palaeohistoria* 35/36: 307-321
- Zeist W van, Woldring H, Neef R (1994) Plant husbandry and vegetation of early medieval Douai, northern France. *Veget Hist Archaeobot* 3: 191-218